James Hun Kim (Junghun James Kim)

AI Researcher & Engineer

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Summary

I am an AI researcher specializing in differentiable rendering, generative models, and 3D vision. My work focuses on building optimization-driven systems that connect research with large-scale production, covering scalable engineering and real-world deployment. At Samsung, I led the end-to-end development of a range of AI-powered camera features, from model design to commercialization, deploying real-time vision systems across smartphones, tablets, and foldable devices under strict latency, memory, and power constraints across diverse hardware environments. I am particularly interested in developing efficient and scalable AI systems that translate advanced research into production.

Work Experience

Samsung Electronics (MX) *Seoul, Republic of Korea*

Visual AI Group *Jan. 2019 – Present*

- Worked on Portrait Video, a real-time AI-based video bokeh system supporting humans, pets, and objects across diverse camera configurations including single, dual, and ToF cameras.
- Led end-to-end development and commercialization.
- Deployed across multiple Galaxy product lines from Galaxy S10 5G to Galaxy S25.
- Developed core modules including detection, depth estimation, segmentation, tracking, and visual effects(Blur, Big Circle, Colorpoint, and Glitch).
- Built MLOps pipelines for large-scale debugging and evaluation, significantly improving development efficiency.

Samsung Electronics (MX) *Seoul, Republic of Korea*

Visual S/W R&D Group *Aug. 2017 – Dec. 2018*

- Worked on Bixby Vision, an AI-based visual search system.
- Managed and scaled production systems for Shopping, OCR, and Landmark services.
- Deployed across multiple Galaxy product lines from Galaxy S8 to Galaxy Note 10.
- Established user-feedback-driven evaluation pipelines and survey frameworks for systematic preference analysis.

Samsung Electronics (MX) *Seoul, Republic of Korea*

SE Team *Jul. 2016 – Aug. 2017*

- Worked on SCM (Software Configuration Management), including version release management, firmware management, and Perforce-based version control workflows.
- Owned the full lifecycle from design through development of an automated release process system.
- Automated the firmware release pipeline, reducing required manpower from about 30 engineers to 3.

Publications

HyFL-CLIP: Hyperbolic Fine-Tuning of CLIP for Robust Long-Context Understanding *ECCV*

Ji Ha Jang*, Hayeon Kim*, Chulwon Lee, Junghun James Kim, Se Young Chun (*co-first authored) *2026*

A hyperbolic fine-tuning framework that distills the well-established image-text alignment of Euclidean CLIP into hyperbolic space via cross-manifold similarity distillation.

Human Interaction-Aware 3D Reconstruction from a Single Image *CVPR, Highlight*

Gwanghyun Kim*, Junghun James Kim*, Suh Yoon Jeon*, Jason Park, Se Young Chun (*co-first authored) *2026*

A holistic framework for reconstructing physically plausible, high-fidelity textured 3D humans from a single image, explicitly modeling group and instance level cues to handle perspective distortion, occlusion, and inter-human interactions.

DiffBMP: Differentiable Rendering with Bitmap Primitives *CVPR*

Seongmin Hong*, Junghun James Kim*, Daehyeop Kim, Insoo Chung, Se Young Chun (*co-first authored) *2026*

A scalable differentiable renderer for bitmap primitives that optimizes thousands of elements via a highly parallelized custom CUDA pipeline, enabling practical image/video composition, layered export, and artist-friendly creative workflows.

Uncertainty-guided Compositional Alignment with Part-to-Whole Semantic Representativeness in Hyperbolic Vision-Language Models *CVPR, Highlight*

Hayeon Kim*, Ji Ha Jang*, Junghun James Kim, Se Young Chun (*co-first authored) *2026*

An uncertainty-guided hyperbolic vision-language framework that models part-to-whole semantic representativeness via adaptive uncertainty, improving hierarchical compositional understanding and performance on zero-shot classification, retrieval, and multi-label classification.

One-Shot Item Search with Multimodal Data *arXiv preprint*
Jonghwa Yim, Junghun James Kim, and Daekyu Shin *2018*
A multimodal item retrieval method that jointly searches over image and text features rather than handling them separately, improving similar-item search on large-scale shopping datasets with only minimal additional computational overhead.

Patents

Electronic device and method for increasing resolution of digital bokeh image *Mar, 2024*
US 20250272806, EP 4604568A4
An image processing method for digital bokeh photography that identifies blur-heavy regions around a user-selected subject and selectively increases local resolution for clearer portrait-style results.

Electronic device and method for generating image *Mar, 2023*
US 2024/0314429, KR 10-2794879
An adaptive camera pipeline that switches between stereo and AI-based bokeh generation according to the subject, balancing image quality with lower power consumption and heat.

Electronic device for providing information on item based on category of item *Nov, 2022*
US 11500918B2, KR 10-2734142
A category-aware visual retrieval method that matches a query item using both image features and hierarchical category labels, then re-ranks results with brand, logo, and attribute cues.

Electronic device, method, and computer-readable medium for providing bokeh effect in video *Feb, 2022*
US 11258962B2, EP 3925206A4, CN 113647094B, KR 10-2835449
A multi-camera video bokeh method that tracks frame-to-frame subject distance and adaptively updates background blur regions to maintain stable, high-quality bokeh rendering in video.

Education

Seoul National University *Seoul, Republic of Korea*
M.S. in IPAI(Interdisciplinary Program in Artificial Intelligence) *Mar. 2026 – Present*

Hongik University *Seoul, Republic of Korea*
B.S. in Computer Science & Engineering *Feb. 2016*

References

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